

000020

LOT 4 : SCHEDULE S4-1

PART-H BOQ FOR Additional Misc. Items

Section No.	DESCRIPTION	Total brought from above Bills	
		INR	FOREIGN CURRENCY USD
PART - H	BOQ FOR Additional Misc. Items	Rs. 3,65,63,350.00	\$86,927.00
TOTAL SUM		Rs. 3,65,63,350.00	\$86,927.00

TOTAL SUM IN WORDS, OF INDIAN RUPEES

Three Crore Sixty Five Lac Sixty Three Thousand Three Hundred Fifty Rupees and No Paise

TOTAL SUM IN WORDS, OF FOREIGN CURRENCY

Eighty Six Thousand Nine Hundred Twenty Seven Dollars and No Cents



000021

LOT 4 : SCHEDULE S4-1

PART-I BOQ FOR AMC beyond DLP

Section No.	DESCRIPTION	Total brought from above Bills	
		INR	FOREIGN CURRENCY USD
PART - I	BOQ FOR AMC beyond DLP	Rs. 6,00,43,530.00	\$0.00
TOTAL SUM		Rs. 6,00,43,530.00	\$0.00

TOTAL SUM IN WORDS, OF INDIAN RUPEES

Six Crore Forty Three Thousand Five Hundred Thirty Rupees and No Paise

TOTAL SUM IN WORDS, OF FOREIGN CURRENCY

No Dollars and No Cents



000027



LOT 4 : SCHEDULE S4-1

PART- J BOQ FOR SARES

Section No.	DESCRIPTION	Total brought from above Bills	
		INR	FOREIGN CURRENCY USD
PART - J	BOQ FOR SPARES	Rs. 2,38,03,130.00	\$3,22,657.00
TOTAL SUM		Rs. 2,38,03,130.00	\$3,22,657.00

TOTAL SUM IN WORDS, OF INDIAN RUPEES

Two Crore Thirty Eight Lac Three Thousand One Hundred Thirty Rupees and No Paise

TOTAL SUM IN WORDS, OF FOREIGN CURRENCY

Three Hundred Twenty Two Thousand Six Hundred Fifty Seven Dollars and No Cents



000028



REVISED PART-K BOQ for Training & Training Manuals

Section No.	DESCRIPTION	Total brought from above Bills	
		INR	FOREIGN CURRENCY
PART - K	BOQ FOR TRAINING MANUALS	631070	NIL
TOTAL SUM		631070	NIL

TOTAL SUM IN WORDS, OF INDIAN RUPEES
Six Lakh Thirty One Thousand Seventy Only

TOTAL SUM IN WORDS, OF FOREIGN CURRENCY
NIL

Confidential



000029



September, 2012



000023

LOT 4 : SCHEDULE S4-1

PART-K BOQ for Training & Training Manuals

Section No.	DESCRIPTION	Total brought from above Bills	
		INR	FOREIGN CURRENCY USD
PART - K	BOQ FOR TRAINING MANUALS	Rs. 6,30,490.00	\$0.00
TOTAL SUM		Rs. 6,30,490.00	\$0.00

TOTAL SUM IN WORDS, OF INDIAN RUPEES

Six Lac Thirty Thousand Four Hundred Ninety Rupees and No Paise

TOTAL SUM IN WORDS, OF FOREIGN CURRENCY

No Dollars and No Cents



000030



LOT 4 : SCHEDULE S4-1

PART-L BOQ for Operation and Maintenance Manuals

Section No.	DESCRIPTION	Total brought from above Bills	
		INR	FOREIGN CURRENCY USD
PART - L	BOQ for Operation and Maintenance manuals	Rs. 5,04,390.00	\$0.00
TOTAL SUM		Rs. 5,04,390.00	\$0.00

TOTAL SUM IN WORDS, OF INDIAN RUPEES

Five Lac Four Thousand Three Hundred Ninety Rupees and No Paise

TOTAL SUM IN WORDS, OF FOREIGN CURRENCY

No Dollars and No Cents



000031



SCHEDULE OF QUANTITIES FOR PART-A Electrical WORKS - CE-10&11 LOT-4(ADDENDUM)

S.No.	Description	Units	IGD Airport	mid shaft	Palam	Dashrath Puri	Debri Mor	Janakpuri West	Total Quantity	Rate in INR	Rate in Foreign Currency	Amount in INR	Amount in Foreign Currency
1	Switch Boards/panels												
	Supply, installation, testing and commissioning of front operated, front / back access, cubicle type, indoor duty, floor / wall / recess / surface mounted (as specified), totally enclosed dust and vermin proof switchboards / panels with minimum Ingress protection classification of IP 54, fabricated from 2 mm thick CRCA sheets with dip coat priming and epoxy powder coated finish. The panels must be suitable for 415 volts 3 phase, 4 wire, 50 Hz, system, must be able to withstand symmetrical fault level of 65 or 50 KA (as specified for various panels) for 1 sec at 415 V and must include all interconnections, earthing and bonding requirements etc. The panels must conform to drawings & standards specified in relevant specifications, as required and as given below.												
a	Main LV Switchboards must conform to Section E01 of the M & W specifications												
b	All Sub - Main LV Switchboards must conform to Section E05 of the M & W specifications												
c	All Local Electrical Panels must conform to Section E06 of the M & W specifications												
d	All the items mentioned in relevant Sections / Clauses of the M & W specifications and not specifically mentioned in BOQ shall be deemed to be included in the quoted rates												
e	In case of any contradiction between BOQ / M & W Specifications and Drawings, the most stringent conditions of the above will apply.												
f	The Switchboards shall be provided with detachable gland plates for entry of cables from the top / bottom as required.												
g	All live accessible parts shall be shrouded and all equipment shall be finger touch proof. SMC / DMC shrouds and busbar supports suitably spaced shall be used. Hinged doors with padlocking facility shall be provided on all outgoing feeders with switch handles lockable in OFF position.												
h	The panel shall have copper busbars as specified in Section E01 of M & W Specifications with bar type feeder connections, spacers etc. with full sized neutral.												
i	Earthing all components, frames etc. to a common internal earth bar of size 50 x 6 mm Cu. or higher, as required by the fault withstand level specified for the board.												
j	All accessories & supporting structures such as channels, ISMC-100 base frame, mounting brackets, lifting lugs, panel heaters, ventilation arrangement etc shall be provided as required.												
k	Each incomer and outgoing feeder shall be provided with LED type status indication lamps suitable for 240 V, AC, as approved.												

SCHEDULE OF QUANTITIES FOR PART-A Electrical WORKS - CE-10 and 11 LOT-4(ADDENDUM)

S.No.	Description	Units	IGD Airport	mid shaft	Palam	Dashrath Puri	Dabri Mor	Janakpuri West	Total Quantity	Rate in INR	Rate in Foreign Currency	Amount in INR	Amount in Foreign Currency
l	Space provision as specified in M & W Specifications for future expansion.												
m	The make of all switchgear (ACB / MCCB / MCB etc.) must be same to have total discrimination.												
n	Components and accessories shall be same for switchboards and panels for uniformity, standardisation and replaceability and shall be applicable to all panels/ boards under the scope of work												
o	All interconnections, labeling, earthing, associated foundation / masonry work, sealing of gaps around the panels by chequered plates & erection etc. complete shall be executed as required.												
p	MCCB's shall be as per Section E01, E05 and E06 of M & W Specifications and shall be current limiting type with front adjustable electronic releases for required protections against fault, suitable for isolation as per Annexure 7.1.2 of IEC 60947 - 2, with Ics and Icu values as specified in relevant clauses of M & W Specifications and conforming to latest IEC 60947. The operating operating voltage (Ue) shall be 415 V and insulation voltage (Ui) 690V with trip free mechanism, handle indicating ON / OFF / tripped positions. The breaking capacity as mentioned in items of BOQ below, shall be Ics values and shall not be less than 56kA. Breakers must have Ics = 100% Icu. Electronic trip units shall comply with the requirements as specified in Appendix F (EMC /EMI Compatibility) of IEC 60947-2 or EN 60947-2.												
q	MCCBs for Network / Feeder Protection shall have adjustable protection against Overload and adjustable protection against Short Circuit as specified in Section E01 and E05 of M & W Specifications, whether specifically asked in BOQ or not. MCCB's for Network / Feeder Protection shall have releases with Earth Fault protection feature, wherever and as indicated in Bill of Quantities or Drawings. However, on completion of Cable Sizing Calculations and Fault calculations by the Contractor, if Earth Fault releases are required for additional feeders, where this requirement is not indicated in the BOQ, same shall be provided at no extra cost to the Employer. For Motor Protection MCCB's, requirements of Section E06 of M & W Specifications shall be followed.												
r	ACBs/MCCBs shall be compact , suitably designed to provide protection of motors, cables, busbars to suit rated current, unbalanced power distribution, as required.												
s	Panel / Switchboard board design shall be compact and components / accessories of compact sizes shall be used to economise the room space available. Employer reserve the right to seek compact items in place of larger ones.												
t	Incomer ACB'S / MCCBs / MCBs of switchboards / panels shall be provided with NO / NC contacts as specified in specifications for interface with SCADA System and for manual / auto and local / remote operation.												

SCHEDULE OF QUANTITIES FOR PART-A Electrical WORKS - CE-10.1 LOT-4(ADDENDUM)

S.No.	Description	Units	IGD Airport	mid shaft	Palam	Dashrath Puri	Dabri Mor	Janakpuri West	Total Quantity	Rate in INR	Rate in Foreign Currency	Amount in INR	Amount in Foreign Currency
u	ACB's shall have Microprocessor based releases as specified in the M & W Specifications.												
v	In case higher frame sizes (than those specified in the BOQ) are required to be provided to achieve the requirement of Total Discrimination mentioned in Section E01 and E05 of M & W Specifications, due to selection of a particular make, the same shall be provided at no extra cost if other makes are able to achieve the same with the specified frame size.												
w	Contractor must carry out Cable Sizing based on actual lengths of cables and Discrimination study for selection of trip units and settings for the releases during production of Working Drawings.												
x	All Motor feeders, having separate Thermal Over - load relay, shall be provided with Reset push button.												
y	Even though, ratings of MPCB / Contactors for Motor Feeders are mentioned in BOQ items below, after Selection of a particular make of Switchgear, MPCB / Contactors with ratings as per tested Type II Coordination from the selected manufacturer only shall be provided.												
	Description of Various panels/boards covered under this BOQ is as given below:												
1.01	(Type-1 for chiller plant with 3 chillers)												
	MAIN PLANT PANEL DB-170/270 (Type-1)												
	Panel complete as per specifications, drawings, requirements & following details:	Sets			1	1	1		3	8810960		26432880	
A	INCOMERS												
1)	2 Nos. 2500A, 415 V, 65 50KA, 50 Hz TPN EDO type ACBs interlocked with buscoupler and complete with microprocessor based releases having SCADA compatibility and energy metering.												
2)	2 sets of Red (on), Green (off) and amber (C.B. Fault) indicating lamps												
3)	2 sets of (three) white phase indicating lamps												
4)	Metering												
i)	2 Nos. Multifunction energy meter for measurement of voltage, line current, power factor, power & energy.												
ii)	3 Nos. CTs 2500/5A, 15VA, CL-1 for metering.												
5)	Terminals to receive XLPE armoured power & control cables												
B	BUSBAR												
	2 Nos. Electrolytic high conductivity tinned copper three phase and neutral busbars rated at 2500 A having a maximum current density of 1.2 A per sqmm suitable to withstand symmetrical fault level of 56 50KA at 415 V.												
C	BUS COUPLER												
1)	1 Nos. 2500A, 415 V, 65 50KA, 50 Hz TPN EDO type ACBs interlocked												
2)	Terminals to receive XLPE armoured power & control cables												
3)	Digital Ammeter with Inbuilt Selector Switch												
4)	3 Nos. CTs, 2500/5A, 15VA, CL-1 for metering.												
D	OUT GOING FEEDERS												
1	3 Sets feeder for Chiller Motors up to 276 KW each shall be equipped with the followings but not limited to:												

SCHEDULE OF QUANTITIES FOR PART-A Electrical WORKS - CE-1 LOT-4(ADDENDUM)

S.No.	Description	Units	IGD Airport	Mid shaft	Palam	Dashrath Puri	Dabri Mor	Janakpuri West	Total Quantity	Rate in INR	Rate in Foreign Currency	Amount in INR	Amount in Foreign Currency
a)	1 No. 800 A, 415V, 65 50kA, 3 Pole, Electrically operated Draw Out type, ACB complete with safety shutters, 240V AC or 110V DC (control supply voltage to be selected in consultation with BMS Contractor) shunt trip coil & closing coil and having microprocessor based releases with (1) Protection against long time + short time + instantaneous + earth fault (all with adjustable time delays) (2) NO / NC contacts for interlocks and indications (3) 1 no CT in neutral to connect with the trip unit.												
b)	TNC switch												
c)	Selector switch for auto/manual operation.												
d)	LED indication lamp for ON, OFF & TRIP.												
e)	Digital Ammeter with Inbuilt Selector Switch												
f)	3 Nos. CTs 800/5A, 15VA, CL-1 for metering.												
g)	Digital Energy meter.												
h)	Power & control terminals for cable connections.												
2	4 Sets of Star - Delta Starter Unit for Primary Chilled Water Pump Motors of 42 15KW (3 Working + 1 Standby) and each shall be equipped with the followings but not limited to:(For -CHWP-001P, CHWP-002P, CHWP-003P, CHWP-04P)												
a)	1 No. 3 Pole 100A, 50kA at 415V, 3P motor duty MCCB with Electronic												
b)	1 No. Digital Ammeter, CT operated.												
c)	1 No. CT 100/5A, 15VA, class - I for metering.												
d)	Local / Remote Selector Switch key operated.												
e)	Working / Standby Selector Switch key operated.												
f)	Door mounted push buttons for Start and Stop functions.												
g)	Emergency stop push button lockable type.												
h)	LED indication lamp for ON, OFF, Trip.												
i)	3 Nos suitable rating AC-3 duty, 3 Pole Contactor with Auxiliary												
j)	One no star-delta timer.												
k)	1 No. Auxiliary Contactor with 4 Nos. Auxiliary Contacts for Remote												
l)	1 No Single Phasing Preventor Relay.												
m)	Power & control terminals for cable connections.												
n)	Digital Energy meter.												
o)	Provision for terminals for wiring to Local control panel (LCP)												
3	3 Nos. Feeder for Secondary Chilled Water Pump Motors of 37 KW (2 Working + 1 Standby) and each shall be equipped with the followings but not limited to:(For CHWP-001S, CHWP-002S, CHWP-003S)												
a)	1 No. 3 Pole 100A, 50kA at 415V, 3P motor duty MCCB with Electronic												
b)	1 No. Digital Ammeter, CT operated.												
c)	1 No. CT 100/5A, 15VA, class - I for metering.												
d)	LED indication lamp for ON, OFF, Trip.												
e)	Digital Energy meter.												
4)	4 Sets of Star - Delta Stater unit for Condenser Water Pump Motors up to 37 KW (3 Working + 1 Standby) and each shall be equipped with the followings but not limited to:(For CDWP-001, CDWP-002, CDWP-003, CDWP-04)												
a)	1 No. 3 Pole 100A, 50kA at 415V, 3P motor duty MCCB with Electronic												
b)	1 No. Digital Ammeter, CT operated.												
c)	1 No. CT 100/5A, 15VA, class - I for metering.												
d)	Local / Remote Selector Switch key operated.												
e)	Working / Standby Selector Switch key operated.												
f)	Door mounted push buttons for Start and Stop functions.												